Similar Areas & Volumes

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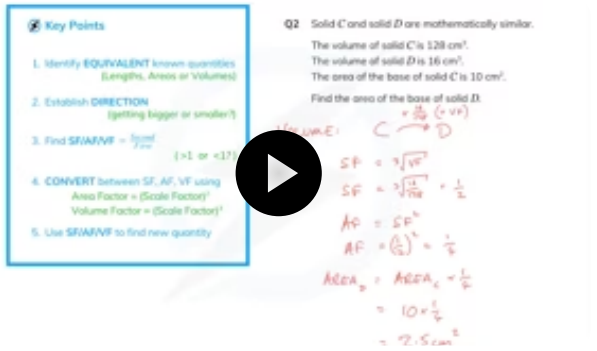

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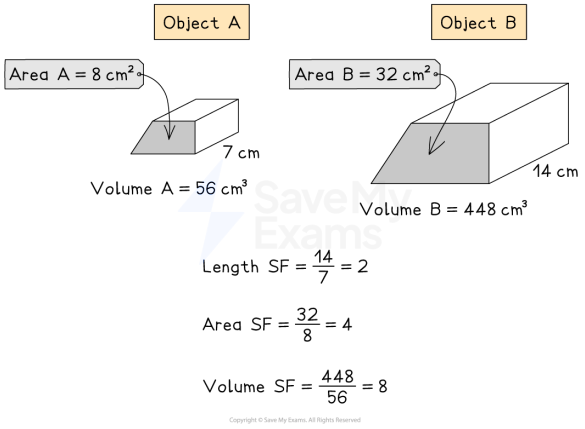
Similar areas & volumes

How do I find the length, area or volume scale factors of similar shapes?

- The **scale factor** (SF) for a given quantity (length, area or volume) between **two similar shapes** can be found by dividing the quantity on one shape by the quantity on the other shape
  - $\text{scale factor} = \dfrac{\text{quantity on one shape}}{\text{corresponding quantity on the other shape}}$



Example of scale factor of similar shapes

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

than 1
    - The object gets **smaller** if the scale factor is **bigger than 0** and **less than 1**

What is the connection between the scale factors for lengths, areas and volumes of similar shapes?

- The **length**, **area** and **volume scale factors** are powers with the same base number
- If the length **scale factor** is k then
    - The **area scale factor** is k^2
    - The **volume scale factor** is k^3

- If you know one scale factor, you can find the **scale factors**
    - If you have the **length** scale factor
        - $\text{Area scale factor} = (\text{Length scale factor})^2$
        - $\text{Volume scale factor} = (\text{Length scale factor})^3$

    - If you have the **area** scale factor
        - $\text{Length scale factor} = \sqrt{\text{Area scale factor}}$
        - $\text{Volume scale factor} = \left(\sqrt{\text{Area scale factor}}\right)^3$

    - If you have the **volume** scale factor
        - $\text{Length scale factor} = \sqrt[3]{\text{Volume scale factor}}$
        - $\text{Area scale factor} = \left(\sqrt[3]{\text{Volume scale factor}}\right)^2$

How do I find missing lengths, areas and volumes for similar shapes?

- **STEP 1**
  Identify the **equivalent** known quantities
    - Recognise if the quantities are lengths, areas or volumes

- **STEP 2**
  Find the **scale factor** from two known **lengths, areas** or **volumes**
    - $\text{scale factor} = \dfrac{\text{second quantity}}{\text{first quantity}}$

- **STEP 3**
  Use the scale factor you have found to find **other required scale factor(s)**
    - $\text{Length scale factor} = k$
    - $\text{Area scale factor} = k^2$
    - $\text{Volume scale factor} = k^3$

- **STEP 4**
  Multiply or divide by relevant scale factor to find the **missing quantity**
    - Think about whether the quantity should be bigger or smaller than the given quantity

have started carrying out the calculations, it can help to **label the shapes** and write an equation.

Worked Example

Solid *A* and solid *B* are mathematically similar.

The volume of solid *A* is 32 cm^3.
The volume of solid *B* is 108 cm^3.
The height of solid *A* is 10 cm.

Find the height of solid *B*.

Answer:

Calculate k^3, the scale factor of enlargement for the volumes, using: $\text{volume } B = k^3(\text{volume } A)$

Or $k^3 = \dfrac{\text{larger volume}}{\text{smaller volume}}$

$$108 = 32k^3$$
$$k^3 = \frac{108}{32} = \frac{27}{8}$$

Find the length scale factor k by taking the cube root of the volume scale factor k^3

$$k = \sqrt[3]{\frac{27}{8}} = \frac{3}{2}$$

Substitute the value for k into formula for the heights of the similar shapes:

$\text{Height } B = k(\text{height } A)$

$$h = 10k$$
$$h = 10\left(\frac{3}{2}\right) = \frac{30}{2} = 15$$

Height of *B* = 15 cm

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